

Measuring ploidy (propidium iodide staining)

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Measuring ploidy (propidium iodide DNA staining)

1. Grow the strain of interest in YPD to mid-log phase; transfer 250 μ L culture to a 1.5 mL tube.
2. Pellet cells ($5,000 \times g$, 1 min), wash with 1 mL H_2O .
3. Macroscopic strains: break clusters into small pieces by vigorous pipetting with 100 μ L regular-bore tips before washing.
4. Fix/permeabilize in 1 mL 70% ethanol, room temperature, 2 h, end-over-end rotation (max speed).
5. Wash twice with 1 mL 50 mM sodium citrate.
6. Resuspend in 200 μ L 50 mM sodium citrate + 0.5 mg/mL RNase A; incubate at 37 °C for 2 h, inverting gently every 30 min.
7. Add 5 μ L of 1 mg/mL propidium iodide; incubate at 30 °C overnight in the dark, gentle rotation (minimum speed).
8. PI-stained clusters may be stored at 4 °C for up to 1 week before imaging.
9. Crush 5 μ L of stained clusters into a single cell layer between slide and coverslip for imaging.
10. Image at $\times 20$ magnification, red fluorescent channel (600 ms exposure, $2.2\times$ gain) at the focal plane plus one z-plane 0.3 μ m above and below (3 planes total, later max-projected), plus a brightfield image at the focal plane (100 ms exposure, $4.1\times$ gain). Set focal plane in brightfield, then move down 1 μ m for nuclear signal.
11. Calibrate exposure/gain so the brightest pixels in a tetraploid control strain are $\sim 80\%$ of maximum pixel value.
12. Run known-ploidy controls (e.g., engineered “grande” diploid and tetraploid snowflake yeast strains) through the same procedure alongside samples in every experiment.